Research Proposal

An Empirical Study of IEEE 802.11 and WPA3 Standards Adoption and Feature Prioritization in Campus and Public Hotspot Deployments

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List of Abbreviations

AP	Access Point
BSS	Basic Service Set
FCC	Federal Communications Commission
GHz	Gigahertz
IEEE	Institute of Electrical and Electronics Engineers
iOS	iPhone Operating System
IT	Information Technology
macOS	Macintosh Operating System
MHz	Megahertz
NIST	National Institute of Standards and Technology
OFDMA	Orthogonal Frequency Division Multiple Access
PMF	Protected Management Frames
RF	Radio Frequency
RFC	Request for Comments
SAE	Simultaneous Authentication of Equals
SSID	Service Set Identifier
WPA2	Wi-Fi Protected Access 2
WPA2	Wi-Fi Protected Access 3

1. Introduction

Campus IT teams face a decision-making challenge: they deploy Wi-Fi networks to support thousands of students and staff but lack clear evidence showing which infrastructure improvements address the problems users experience daily, namely slow file uploads, dropped video calls, and security vulnerabilities.

Without empirical guidance, IT operators must guess at upgrade priorities, often relying on vendor marketing claims rather than independent evidence.

This proposed study will measure how specific Wi-Fi features affect real user outcomes in controlled environments that mirror campus and public hotspot settings, producing a prioritized deployment checklist that enables IT operators to invest limited budgets strategically based on measurable impact.

1.1 Research Problem and Significance

This proposal frames the problem around user experience, evidence gaps, and operational impact.

The Human Problem: What Actually Happens

It is 2:30 PM on a Tuesday. A student is sitting in the university library trying to submit a cloud-hosted assignment that is due in one hour. Her laptop is connected to Wi-Fi, but the upload stalls repeatedly- at 45%, at 82%, finally at 93%. She loses the connection, has to start over, and watches the clock. By the time she resubmits, the assignment is marked late.

In the dormitory next door, another student is in an online group study session via video call with classmates in three different time zones. As she walks from her room to the common area, the call drops. She reconnects, but the conversation thread is broken, and the group loses momentum.

In the campus café, a third student opens their device and finds it has been disconnected she later learns that hackers on the same network broadcast a simple attack that forced devices offline.

These are not rare incidents. IT staff fielding help desk tickets know this: when students complain about Wi-Fi, they are not complaining about technology, they are complaining about lost time, disrupted learning, and broken trust in the institution's infrastructure.

A campus IT manager reviewing budget requests faces a parallel frustration: vendors offer Wi-Fi upgrades with promises of "better performance" and "enhanced security," but the documentation is dense with technical jargon. When the manager asks, "Will this actually help my users?" the honest answer from most vendors is unclear. The IT team has no way to know which improvements matter most or which are worth the budget spent.

The Core Problem: Missing Evidence

IT operators lack concrete evidence showing which Wi-Fi improvements help their users. Vendors claim that features like advanced congestion management, faster roaming, and stronger security will improve performance, but these claims are rarely tested independently in real campus or public-space environments.

Without evidence-based guidance, IT teams guess at priorities, postpone upgrades due to unclear return on investment, and users experience preventable service disruptions.

Why This Matters

This problem is significant for three interconnected reasons.

First, Wi-Fi has become foundational to education and daily life: students rely on it to access cloud-hosted learning platforms, participate in hybrid classes, and collaborate across distances. When Wi-Fi fails, education is disrupted.

Second, the costs of inaction are real; wasted staff time on troubleshooting, users losing confidence in the institution, and budget spent reactively instead of strategically.

Third, Wi-Fi standards are evolving rapidly, and resource-constrained campus IT teams (and small businesses running public hotspots) lack the empirical guidance needed to prioritize which improvements deliver the most value. Without evidence-based recommendations, they default to expensive "solutions" that may not address the actual problems their users face.

This proposed study will bridge that gap by providing clear, measurable evidence of what works and in what priority order, enabling operators to invest budgets wisely and align upgrades with institutional IT priorities.

1.2 Research Objective

This proposed study will test specific Wi-Fi features in controlled laboratory settings and measure whether they improve the outcomes students and staff care about: faster uploads, stable video calls, and resilience to security attacks. This will be achieved by answering the following research questions:

1.3 Research Questions

1. Which Wi-Fi features most improve upload speed, reduce call drops during movement, and provide stable connectivity in crowded campus settings?
2. How much better does newer Wi-Fi security protect against common attacks compared to older standards?
3. What features deliver the highest return on investment for IT operators with limited budgets?

The answers will be used to provide a prioritized deployment checklist that enables IT operators to invest limited budgets strategically based on measurable feature impact, filling the gap between technical standards documentation and operational practice.

2. Literature Review

Existing research shows that Wi-Fi features can improve performance in controlled tests (IEEE, 2020, 2021), but campus IT teams lack practical, independent evidence about which features help their specific users. Published studies tend to examine individual features in isolation and do not address the operational question IT managers face: "If I have a limited budget, which Wi-Fi improvements should I invest in first?"

Prior work demonstrates that spatial reuse features such as BSS coloring and OFDMA improve throughput in dense deployments (IEEE, 2021), and fast roaming standards including IEEE 802.11r reduce handoff latency for voice and video applications (IEEE, 2008a, 2008b, 2011). However, no published studies compare roaming performance across multiple vendor implementations in multi-building campus environments where client devices roam between different access point brands -a common real-world constraint that produces inconsistent performance despite standards compliance.

Security research highlights that WPA2 networks without Protected Management Frames remain vulnerable to deauthentication attacks (Harkins, 2015; National Institute of Standards and Technology, 2012), while WPA3 with SAE and mandatory Protected Management Frames mitigates these attacks (Wi-Fi Alliance, 2024). However, empirical evidence quantifying the real-world security impact in operational campus environments where legacy devices must coexist with modern infrastructure is absent.

Emerging research on 6 GHz spectrum allocation and Wi-Fi 7 (802.11be) projects lower latency and higher capacity (Federal Communications Commission, 2020; IEEE 802.11 Working Group, 2024; Wi-Fi Alliance, 2023). Campus operators lack guidance on when these next-generation features will deliver sufficient return on investment.

Research Gap: No published research has systematically evaluated congestion management, roaming, and security features together in realistic campus and public-venue environments to provide prioritized deployment guidance. Existing studies examine features in isolation, use homogeneous test configurations, and report technical metrics rather than user-visible outcomes. This study bridges the gap by testing multiple features with heterogeneous device populations, measuring real user outcomes (upload speed, call continuity, security resilience), and providing actionable deployment priorities.

3. Research Design

This section details the planned study design and methods, following established empirical research design frameworks that structure investigations through setting, methods, instruments, sampling, analysis, synthesis, validation, and constraints (Hemmer & Fröhlich, 2025).

3.1 Study Type and Setting

This research proposal will use laboratory testbeds that mimic three realistic scenarios: (1) crowded campus spaces like lecture halls and libraries where many devices compete for bandwidth, (2) a multi-building campus where users move around and need stable connections, and (3) a public Wi-Fi hotspot where security threats are a concern. Laboratory studies enable controlled environmental conditions where variables can be systematically manipulated to isolate feature impacts (Hemmer & Fröhlich, 2025).

Figure 1 will illustrate the three laboratory environments and their respective Wi-Fi infrastructure topology, depicting each testbed planned to replicate real-world deployment contexts.

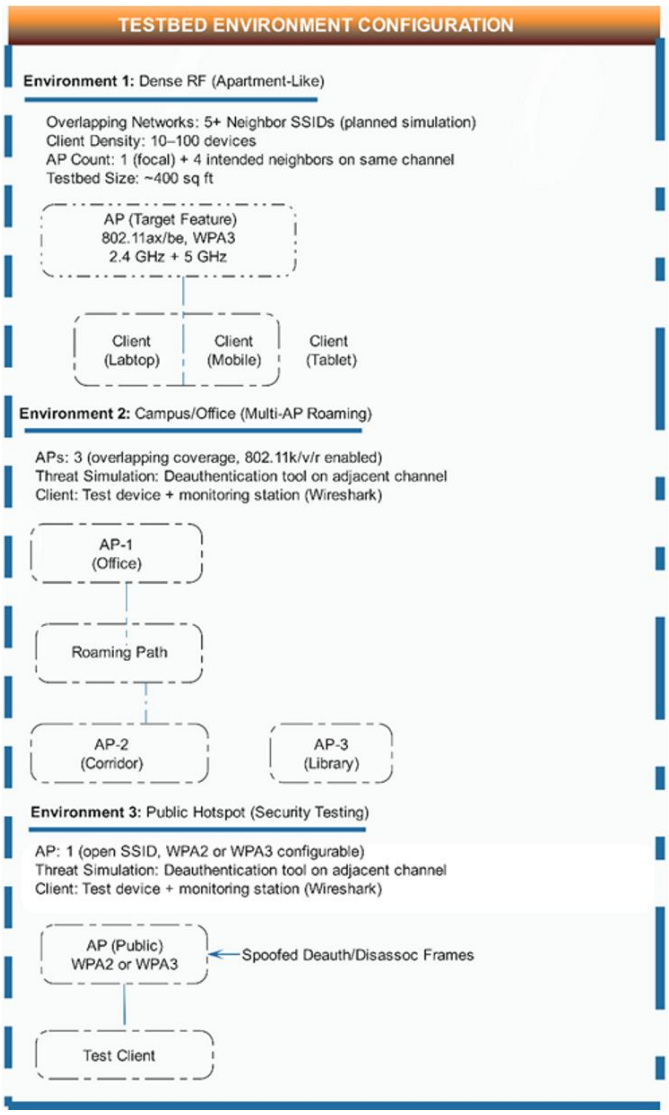

Figure 1. *Testbed Environment Layout and Configuration (Planned)*

Environment 1 will simulate high-density scenarios common in lecture halls and dormitories where co-channel interference and congestion degrade performance. Environment 2 tests anticipated real-world roaming behavior across a campus-like multi-AP deployment. Environment 3 will isolate security threats on a standalone hotspot.

This multi-environment approach ensures findings generalize across the three primary deployment contexts addressed by the research proposal: congestion management (Environment 1), session continuity (Environment 2), and security resilience (Environment 3).

Each testbed will run independently to avoid cross-environment interference, with environments isolated by time slot (weeks 1-2 setup, weeks 3-4 congestion focus, weeks 4-5 roaming focus, weeks 5–6 security focus).

3.2 Methods and Test Scenarios

The proposed study will test four scenarios to measure whether specific Wi-Fi features improve user outcomes. Following IEEE standards and guidance from security research literature (Harkins, 2015; National Institute of Standards and Technology, 2012; IEEE, 2020, 2021), the study design will isolate feature impacts through controlled toggling:

1) **Crowded Network Scenario:** Test how features for managing network congestion improve upload speed and video streaming stability when many users are connected, following OFDMA and spatial reuse measurement protocols established in IEEE 802.11ax testing (IEEE, 2021).
2) **Roaming Scenario:** Test how faster roaming features maintain video call continuity when users move between access points across campus, measuring handoff behavior against IEEE 802.11r/k/v specifications (IEEE, 2008a, 2008b, 2011).
3) **Security Scenario:** Compare newer and older Wi-Fi security standards to measure protection against common attack types, evaluating WPA2 and WPA3 implementations (Wi-Fi Alliance, 2024) against established security assessment methods.
4) **Future Technology Outlook:** Model projected benefits from next-generation Wi-Fi features using published technical data (Federal Communications Commission, 2020; IEEE 802.11 Working Group, 2024; Wi-Fi Alliance, 2023) (no live testing required).

Each scenario will run minimum 10 test trials to ensure consistent, reliable results.

Table 1 will show the four test scenarios, which Wi-Fi features will be tested in each, and what user-visible outcomes will be measured.

Table 1. *Test Scenarios and Measurements (Planned)*

Scenario	Environment	Features To Be Tested	What Will Be Measured	Trial Count
Crowded Network	Dense spaces (lecture halls, libraries)	Advanced congestion management	Upload speed, video streaming quality, connection stability	10 trials × 4 device counts
Roaming	Multi-building campus	Faster roaming features	Video call continuity, connection drops during movement	10 trials × 3 access points
Security	Public hotspot	Newer security standards (WPA3 vs. WPA2)	Attack success rate, recovery time after attack	10 trials per configuration
Future Technology	Analytical model (no live testing)	Next-generation Wi-Fi (Wi-Fi 7, 6 GHz)	Projected performance gains	Modeling only

Each scenario will test specific features by toggling them on and off to measure their impact. The "ON/OFF" approach will create paired datasets that show exactly how much each feature improves outcomes.

By testing features separately in controlled scenarios, the study will isolate the impact of each improvement and demonstrate which ones help users versus which are just marketing claims.

Multiple trials per scenario ensure results are consistent and not due to random variation. Cross-vendor testing (minimum two manufacturers) will verify findings work across different equipment brands.

3.3 Instruments and Data Collection

The proposed study will use standard tools to measure performance and security:

- Network monitoring tools to track upload speed, latency, packet loss, and call quality
- Traffic generation tools to create realistic user load (web browsing, video streaming, file transfers)
- Security testing tools to evaluate attack resistance
- Configuration controls to enable and disable specific Wi-Fi features

Data collection will follow a 12-week timeline: weeks 1-2 testbed setup, weeks 3-6 experiments, weeks 7-8 analysis, weeks 9-12 reporting. Testing will use both enterprise and consumer-grade access points and client devices (laptops, smartphones, tablets) from at least two vendors to ensure findings are not limited to a single manufacturer.

3.4 Sampling Procedure

Testing will include enterprise-grade and consumer-grade access points with documented support for modern Wi-Fi features, and client devices spanning Windows, macOS, iOS, and Android systems. This mix reflects real-world campus environments where users have diverse devices.

Table 2 will specify the types of equipment and devices that will be used for testing to ensure findings apply across different manufacturers and operating systems.

Table 2. *Equipment and Device Coverage (Planned)*

Category	Type 1	Type 2	Features Supported	Purpose
Access Points	Enterprise-grade (e.g., Cisco, Aruba)	Consumer-grade (e.g., Netgear, TP-Link)	Modern Wi-Fi standards, security features, roaming support	Test if findings work on both high-end and budget equipment
Laptops	Windows 11 (Intel Wi-Fi)	macOS (Apple Silicon)	Full modern Wi-Fi support	Represent typical student/ staff devices
Smartphones	Android (Samsung/ Google)	iOS (iPhone)	Modern Wi-Fi and security	Most common mobile devices on campus
Tablets	iPad (Wi-Fi)	Windows Surface	Modern Wi-Fi support	Additional device category for completeness

Using two different manufacturers/platforms per category will ensure results are not specific to one brand. This multi-vendor approach will test whether features work consistently across the diverse equipment mix found in real campus environments.

Campus IT teams manage networks with hundreds of different device types. Testing across multiple manufacturers will provide confidence that recommendations will work for their actual users, not just in ideal laboratory conditions.

All test devices will be running current operating system versions (within 2 major releases) to reflect what students and staff use. This will ensure findings remain relevant for current deployment decisions.

3.5 Analysis Techniques

Findings will be analyzed using the following statistical methods:

- **Comparative analysis:** Paired t-tests will compare performance with features enabled versus disabled within the same environment ($\alpha = 0.05$). One-way ANOVA with post-hoc Tukey tests will identify significant feature differences across multiple conditions.
- **Effect size:** Cohen's d will distinguish statistically significant but trivial improvements ($d < 0.2$) from meaningful performance gains ($d \geq 0.5$).
- **Sample size:** The minimum 10 trials per scenario will be determined through power analysis to detect a 15% performance difference with 80% statistical power ($\beta = 0.20$) at $\alpha = 0.05$, assuming coefficient of variation ≤ 20%.
- **Results presentation:** Box plots, before-after charts, and 95% confidence intervals will show how features improve user-visible outcomes (upload speed, call continuity, security

resilience). Results will be presented separately for each deployment scenario to reflect real-world usage patterns.

All findings will use plain language suitable for IT operators and decision-makers.

3.6 Approach to Synthesis

Findings will be synthesized into a prioritized deployment checklist for IT operators. High-priority features will be those that show measurable improvement in user outcomes (faster uploads, fewer call drops, better security) with reasonable implementation costs. Results will be organized by deployment context (crowded networks, roaming, security) to help operators prioritize based on their specific environment.

3.7 Validation and Reliability

To ensure reliability of methods and implement countermeasures against data bias (Hemmer & Fröhlich, 2025), proposed strategies will include:

- **Repeatability:** Minimum 10 trial runs per scenario; variance and confidence intervals will be reported.
- **Cross-vendor validation:** Sanity checks will use at least two AP and two client vendors to ensure findings will not be vendor-specific.
- **Environmental logging:** Radio frequency conditions (channel utilization, interference levels) will be logged for each trial to control confounds.
- **Protocol compliance:** Configuration will be verified through packet captures to confirm features are genuinely enabled.

3.8 Methodological Constraints

The proposed study acknowledges the following limitations of the selected methodology and methods (Hemmer & Fröhlich, 2025):

- **Hardware availability:** Wi-Fi 7 (802.11be) hardware will not be widely available; 6 GHz projections will rely on analytical modeling with published parameters rather than direct measurement.
- **Vendor firmware variability:** Feature implementation quality will vary across vendors; findings may not generalize to all equipment.
- **Laboratory versus production:** Controlled testbed may not capture all real-world variables (e.g., device roaming behavior, application-layer optimizations).
- **Time and resource limits:** 12-week timeline will constrain depth of testing; focus will be on most impactful features.

3.9 Ethical Constraints

This research proposal will observe the following safeguards:

- **Network ownership:** All security tests will be conducted exclusively on owned laboratory equipment; no testing on third-party networks.
- **Data privacy:** No personal data will be collected; all traffic will be synthetic (iperf, video streaming test patterns).
- **Responsible disclosure:** Any novel vulnerabilities discovered will be disclosed to vendors before publication per coordinated disclosure practices.
- **Environmental impact:** Equipment will be reused where possible; e-waste minimized through responsible disposal.

4. Expected Outcomes

This section describes the types of evidence this proposed study will generate and how findings will support decision-making.

4.1 Value for Decision-Makers

This proposed study will provide:

1) **Practical Evidence on Feature Impact:** Measurement of how specific Wi-Fi features improve the user-visible outcomes that students and staff actually care about, such as faster uploads, fewer call drops, and better security including across realistic campus and public-venue environments. This evidence will help IT operators and budget decision-makers understand which upgrades deliver real value.
2) **Context-Specific Guidance:** By testing in multiple realistic environments (crowded spaces, multi-building roaming, public hotspots), the proposed study will help operators understand which features matter most in their specific deployment context rather than providing one-size-fits-all recommendations.
3) **Actionable Deployment Checklist:** Clear prioritization guidance showing which upgrades to implement first, how much each matters, and what implementation constraints might exist will be presented in language suitable for IT operators, facility managers, and budget decision-makers, not just technical specialists.

Findings will distinguish between measurements from live testing and projections for emerging technologies, ensuring decision-makers understand the confidence level in each recommendation.

4.2 Stakeholder Value

This research proposal will provide useful evidence to multiple groups:

- **IT Operators:** Clear, measurable evidence of which Wi-Fi improvements actually help their users, enabling budget-smart infrastructure decisions.
- **Campus Decision-Makers:** Business-case justification for Wi-Fi upgrades, showing return on investment and helping prioritize limited budgets.

- **Standards Organizations:** Feedback on which Wi-Fi features deliver practical value in real-world deployment contexts.
- **Vendors:** Insight into which features matter most to operators and where implementation complexity or device compatibility creates barriers.

By investigating multiple features in realistic scenarios, this proposed research will advance practical knowledge for Wi-Fi infrastructure planning in educational and public-access environments.

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